

CURRICULUM VITAE - DR. JAY V. KALINANI

PERSONAL AND CONTACT INFORMATION

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RESEARCH INTERESTS

Numerical relativity, gravitational waves, magnetohydrodynamics, compact binary mergers, multi-messenger astrophysics, relativistic jets, radiation transport, r-process nucleosynthesis, large language models

PROFESSIONAL APPOINTMENTS

AUG 2025 - PRESENT	Post-Doctoral Researcher , University of Illinois, Urbana-Champaign, USA Advisor: Prof. Helvi WITEK
JAN 2023 - JUL 2025	Post-Doctoral Researcher , CCRG, Rochester Institute of Technology, USA Advisor: Prof. Manuela CAMPANELLI
JAN 2022 - DEC 2022	Post-Doctoral Researcher , INAF-Padova, Italy Advisor: Dr. Riccardo CIOLFI

EDUCATION

OCT 2018 - DEC 2021	Ph.D. in Physics , University of Padova, Italy (Summa cum laude) Thesis: <i>Binary Neutron Star Mergers in the Era of Multi-Messenger Astrophysics</i> Advisors: Dr. Riccardo CIOLFI, Prof. Jean Pierre ZENDRI
OCT 2016 - SEP 2018	M.S. in Astrophysics , <i>AstroMundus Programme</i> (110/110 cum laude) Thesis: <i>GRMHD Simulations of Binary Neutron Star Mergers</i> Advisors: Dr. Riccardo CIOLFI, Prof. Viviana FAFONE
SEP 2014 - SEP 2016	M.S. in Applied Solid State Physics , M.S.R.A.S, India (CGPA: 8.17/10) Thesis: <i>Acoustic Scattering by Different Geometrical Structures</i> Advisors: Prof. Michel CASTAINGS, Prof. M. R. SRINIVASAN
JUL 2010 - MAY 2014	B.Tech. in Nuclear Science & Engineering , SRM University, India (CGPA: 9.01/10) Thesis: <i>Design Aspects of a Sodium Heated Steam Generator</i> Advisor: Prof. Ganeshan VAIDAYANATHAN

AWARDS AND SCHOLARSHIPS

JUL 2026 - AUG 2026	Argonne Training Program on Extreme-Scale Computing : focus on key skills, approaches and tools to design, implement and execute applications on exa-scale supercomputers
SEPT 2025 - APR 2026	Scholarship & Teaching for Engineering Postdocs Program : focus on developing teaching, mentoring & research skills to prepare postdocs for faculty positions (USD 750)
OCT 2018 - DEC 2021	CARIPARO PhD scholarship awarded by the Cariparo Foundation, Italy (EUR 99.8K)
APR 2019	Best PhD student talk , PHAROS conference, Playa de Aro, Spain (EUR 100)
OCT 2016 - SEP 2018	Consortium student scholarship , AstroMundus program (EUR 43.4K)
OCT 2016	Best master thesis award , M.S. Ramaiah University, India
APR 2016 - JUL 2016	CEFIPRA research grant , Aquitaine-Karnataka pre-PhD program (EUR 4.5K)
JUL 2010 - JUN 2012	Department scholarship for obtaining Rank 1 in exams, SRM University, India (EUR 2.75K)
JUL 2007 - JUN 2008	AFS (YES) intercultural exchange program scholarship for one year of high school exchange in San Diego, California, USA (~ USD 4K)

RESEARCH GRANTS

2023 - 2025	co-I of the three-year project “JetOut – Breakout, propagation, and observational signatures of short gamma-ray burst jets from binary neutron star mergers”, selected among the top-relevance Italy-USA scientific cooperation projects (2023–2025) and funded by MAECI (Italian Ministry of Foreign Affairs and International Cooperation) on a yearly basis (PI: Riccardo Ciolfi). (EUR 131k)
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COMPUTATIONAL GRANTS

- Total 27 grants as PI or Co-I: >50.4 Million CPU-hours, >1 Million GPU-hours

- **PI of 10 grants** (listed below): >15.4 Million CPU-hours on TACC, CINECA and EuroHPC clusters, 0.64 Million GPU-hours on Frontier cluster

AUG 2026 - FEB 2027	<i>Principal Investigator (PI)</i> for ALCF Director's Discretionary project on cluster Polaris (USA) Project title: <i>Scaling and Validation of Black-Hole Cluster Simulations</i> (5 Thousand GPU-hours)
MAR 2026 - MAR 2027	<i>Principal Investigator (PI)</i> for ACCESS Discover project on DeltaAI (USA) Project title: <i>GPU Validation and Benchmarking of AsterX/CarpetX with CocalIDX Reader for Rotating Neutron Star and Binary Neutron Star Initial Data</i> (11 Thousand GPU-hours)
OCT 2025 - OCT 2026	<i>Principal Investigator (PI)</i> for OLCF Director's Discretionary project on cluster Frontier (USA) Project title: <i>Benchmarking AsterX and CanudaX with magnetized neutron star mergers</i> (160 Thousand GPU-hours)
FEB 2025 - FEB 2026	<i>Principal Investigator (PI)</i> for TACC's Pathways project on Frontera cluster (USA) Project title: <i>Relativistic jet formation & role of bulk viscosity in magnetized neutron star mergers</i> (12.6 Million CPU-hours)
MAY 2023 - MAY 2025	<i>Principal Investigator (PI)</i> for OLCF Director's Discretionary project on cluster Frontier (USA) Project title: <i>Benchmarking for the new GPU-accelerated open-source GRMHD code AsterX</i> (320 Thousand GPU-hours)
DEC 2022 - JUN 2023	<i>Principal Investigator (PI)</i> for MoU INAF-CINECA Class B project on cluster Galileo 100 (Italy) Project title: <i>Binary neutron star mergers with the general relativistic MHD code Spritz and the APR4 equation of state</i> (~375 Thousand CPU-hours)
JULY 2022 - DEC 2022	<i>Principal Investigator (PI)</i> for EuroHPC Benchmark Access project on Discoverer (Bulgaria) Project title: <i>Binary neutron star merger simulations with the GRMHD code Spritz</i> (~1 Million CPU-hours)
MAR 2022 - DEC 2022	<i>Principal Investigator (PI)</i> for CINECA-Class C project on Galileo 100 (Italy) Project title: <i>Magnetized binary neutron star merger simulations with the GRMHD code Spritz</i> (~100 Thousand CPU-hours)
AUG 2021 - MAY 2022	<i>Principal Investigator (PI)</i> for CINECA-Class C project on Galileo 100 (Italy) Project title: <i>Fishbone-Moncrief disk simulations with the Spritz code</i> (~192 Thousand CPU-hours)
APR 2020 - APR 2021	<i>Principal Investigator (PI)</i> for CINECA-Class B project on Marconi 100 and Galileo (Italy) Project title: <i>Binary neutron star merger simulations with Spritz code</i> (~1.25 Million CPU-hours)

SOFTWARE DEVELOPMENT

I lead the development of several high-performance astrophysical codes and contribute to multiple community projects, especially within the Einstein Toolkit ecosystem. My work spans diverse software infrastructures—from AMReX to Cactus—bridging GPU acceleration, GRMHD physics and radiation transport. I strongly advocate for open-source development and collaborative practices that promote transparency, reproducibility, and accessibility in computational astrophysics. Below, I list some of the open-source software I have worked on:

2026 - PRESENT	BHClusterX : Initial-data code for black-hole cluster simulations with the Einstein Toolkit <i>Lead developer & maintainer</i>
2026 - PRESENT	FlowTracerX : GPU-accelerated passive fluid-particle tracer for AsterX <i>Lead developer & maintainer</i>
2026 - PRESENT	ParticleAHFinderX : GPU-accelerated apparent-horizon finder using AMReX particles <i>Lead developer & maintainer</i>

- 2025 - PRESENT **AsterAI**: AI-assisted code review toolkit for numerical relativity
Lead developer & maintainer
- 2024 - PRESENT **nuX**: A new GPU-accelerated moment-based neutrino radiation transport code
Lead developer & maintainer
- 2022 - PRESENT **AsterX**: A new GPU-accelerated GRMHD code for dynamical spacetimes
Lead developer & maintainer
- 2020 - PRESENT **Spritz**: A GRMHD code with neutrino leakage for fully dynamical spacetimes
Developer
- 2022 - PRESENT **SpacetimeX**: A collection of modules to initialize and evolve spacetime
Developer
- 2022 - PRESENT **CarpetX**: A new Cactus driver for the Einstein Toolkit based on AMReX
Contributor
- 2021 **RePrimAnd**: Library for recovery of primitive variables and EOS framework for GRMHD codes
Contributor

PUBLICATIONS

Total 18 journal articles, > 550 citations, h-index 12 (NASA ADS).

Major contributions

- 2026 J.V. Kalinani, M. Campanelli, M. Chabanov, M.C. de Simone, C.O. Lousto, H. Witek, & Y. Zlochower, submitted to ApJL, [arxiv:2609.24837](#)
Jet variability in small mass ratio supermassive black hole binaries
- 2026 F.H. Nouri, J.V. Kalinani, B. Giacomazzo, R. Ciolfi, & A. Perego, submitted to CQG, [arxiv:2608.23310](#)
Binary Neutron Star Merger Simulations with Microphysical Equation of State using Spritz
- 2026 A. Wen, J.V. Kalinani, M. Chabanov, M. Campanelli, R. Ciolfi, & Y. Zlochower, accepted in PRD, [arxiv:2605.30548](#)
Magnetic eruption and nucleosynthesis in GRMHD simulations of spinning neutron star mergers
- 2026 J.V. Kalinani, R. Ciolfi, M. Campanelli, B. Giacomazzo, A. Pavan, A. Wen & Y. Zlochower, [ApJL 1000 L35](#)
Jet-environment interaction after delayed collapse in binary neutron star mergers
- 2024 J.V. Kalinani, L. Ji, L. Ennoggi, F.G.L. Armengol, L.T. Sanches, B.J. Tsao, S.R. Brandt, M. Campanelli, R. Ciolfi, B. Giacomazzo, R. Haas, E. Schnetter & Y. Zlochower, [CQG 42, 025016](#)
AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes
- 2023 A. Pavan, R. Ciolfi, J.V. Kalinani, & A. Mignone, [MNRAS 524:1, 260](#)
Jet-environment interplay in magnetized binary neutron star mergers
- 2022 J.V. Kalinani, R. Ciolfi, W. Kastaun, B. Giacomazzo, F. Cipolletta, & L. Ennoggi, [PRD 105, 103031](#)
Implementing a new recovery scheme for primitive variables in the GRMHD code Spritz
- 2021 A. Pavan, R. Ciolfi, J.V. Kalinani, & A. Mignone, [MNRAS 506:3, 3483](#)
Short gamma-ray burst jet propagation in binary neutron star merger environments
- 2021 F. Cipolletta, J.V. Kalinani, E. Giangrandi, B. Giacomazzo, R. Ciolfi, L. Sala, & B. Giudici, [CQG 38, 085021](#)
Spritz: General Relativistic Magnetohydrodynamics with Neutrinos
- 2021 W. Kastaun, J.V. Kalinani, & R. Ciolfi, [PRD 103, 023018](#)
Robust Recovery of Primitive Variables in Relativistic Ideal Magnetohydrodynamics
- 2020 R. Ciolfi & J.V. Kalinani, [ApJL 900 L35](#)
Magnetically driven baryon winds from binary neutron star merger remnants and the blue kilonova of August 2017
- 2020 F. Cipolletta, J.V. Kalinani, B. Giacomazzo & R. Ciolfi, [CQG 37, 135010](#)
Spritz: a new fully general-relativistic magnetohydrodynamic code
- 2019 R. Ciolfi, W. Kastaun, J.V. Kalinani & B. Giacomazzo, [PRD 100, 023005](#)
First 100 ms of a long-lived magnetized neutron star formed in a binary neutron star merger
- 2014 J.V. Kalinani & A.R. Kumar, [IJRSET, vol.3, 11, no.68](#)
Thermal Analysis of a Mobile Hot Cell Cask

Minor contributions

- 2026 L. T. Sanches, S. R. Brandt, J. V. Kalinani, L. Ji & E. Schnetter, [CQG 43, 065010](#)
Accelerating numerical relativity simulations with new multi-step fourth order Runge-Kutta methods
- 2025 L. Ennoggi, M. Campanelli, Y. Zlochower, S.C. Noble, J. Krolik, F. Cattorini, J.V. Kalinani, V. Mewes, M. Chabanov, L. Ji & M.C. de Simone, [PRD 112, 063009](#)

Relativistic gas accretion onto supermassive black hole binaries from inspiral through merger

2025 A. Pavan, R. Ciolfi, E. Dreas, J.V. Kalinani, [MNRAS 540, 1345–1362](#)

Role of injection parameters in jet propagation through realistic binary neutron star merger environments

2022 C. Palenzuela, R. Aguilera-Miret, F. Carrasco, R. Ciolfi, J.V. Kalinani, W. Kastaun, B. Minano, & D. Vigano, [PRD 106, 023013](#)

Turbulent magnetic field amplification in binary neutron star mergers

MAIN SCIENTIFIC RESULTS

- 2026 Led three-dimensional simulations of magnetized gas around a precessing, unequal-mass supermassive black-hole binary, tracing its relativistic jet over large distances. Found that the primary black hole powered the jet, while the secondary's repeated disk crossings added matter to the jet and drove quasi-periodic outflows. Identified jet precession, changes in magnetic-field orientation, and periods of jet weakening as connected electromagnetic signatures of the binary. (*Kalinani et al., 2026*)
- 2026 Contributed to the first binary neutron star merger simulations performed with Spritz using a realistic, temperature-dependent nuclear equation of state. Found that the equal-mass merger amplified its magnetic field more efficiently and began launching a polar outflow, while the unequal-mass merger ejected more neutron-rich material but retained dense matter above the poles. Demonstrated a trade-off between conditions favorable for jet formation and the production of kilonova ejecta. (*Nouri et al., 2026*)
- 2026 Contributed to three-dimensional general-relativistic simulations with neutrino transport showing how neutron-star spin shapes merger outflows and element formation. Found that spins aligned with the orbit produced an earlier, more focused magnetic outflow, while anti-aligned spins reduced magnetic-field amplification. Showed that neutrino interactions changed the composition of the ejected material and could produce a distinct late-time kilonova signal. (*Wen et al., 2026*)
- 2026 Performed GRMHD simulations of BNS mergers producing two sequential jets—first powered by the magnetar remnant and later by the BH-disk system consistent with a GRB. Demonstrated that these outflows can interact, generating shock heating and potential electromagnetic signatures, captured self-consistently for the first time. Implemented an unprecedentedly low density floor, effectively eliminating artificial-atmosphere limitations. (*Kalinani et al., 2026*)
- 2025 Contributed to the analysis and interpretation of a new 3D GRMHD framework combining multiple numerical codes to model the inspiral and merger of supermassive black hole binaries with radiative cooling. Simulated an equal-mass, nonspinning binary for ~ 200 orbits, capturing disk evolution, accretion decline, and minidisk disruption through merger. Found a transient luminosity enhancement ($\sim 50\%$) and spectral change at merger, offering potential electromagnetic counterparts to future gravitational-wave detections. (*Ennoggi et al., 2025*)
- 2024 Led the development of AsterX, an open-source, GPU-accelerated GRMHD code for dynamic spacetimes in 3D Cartesian coordinates, built on CarpetX/AMReX and optimized for exascale computing. Implemented the Valencia formulation coupled with the Z4c spacetime evolution and high-resolution shock-capturing schemes. Demonstrated excellent accuracy and efficiency in extensive benchmarks, scaling up to 4096 nodes on OLCF's Frontier supercomputer. (*Kalinani et al., 2024*)
- 2021-2025 Contributed to the hand-off procedure and analysis of the first 3D special-relativistic (M)HD simulations of GRB jet piercing through a realistic magnetized post-merger environment, where the initial conditions were set by importing the data from GRMHD BNS merger simulation (*Pavan et al., 2021, 2023, 2025*)
- 2020-2022 Co-led the development of the GRMHD code Spritz, performed extensive testing, also for the neutrino leakage scheme (*Cipoletta et al., 2020, 2021*). Contributed to the development and testing of the robust primitive variable recovery scheme RePrimAnd (*Kastaun et al., 2021*), which is now widely adopted in various GRMHD codes, and successfully implemented this scheme in Spritz (*Kalinani et al., 2022*)
- 2019–2020 Contributed to GRMHD simulations of BNS mergers forming long-lived neutron star remnants, offering new insights into magnetic field amplification, structural evolution, remnant dynamics, and the conditions for powering a short GRB with or without collapse to a black hole. (*Ciolfi et al., 2019*) Conducted the first detailed study of mass ejection via a magnetically driven, baryon-loaded wind from a massive neutron star remnant, demonstrating that it provides a viable explanation for the “blue” kilonova component of AT2017gfo associated with GW170817. (*Ciolfi & Kalinani, 2020*)

PROFESSIONAL ACTIVITIES

- Local and scientific organizer of the North American Einstein Toolkit Workshop and School 2026, held at UIUC
- Local organizer of the CSSI Einstein Toolkit hackathon 2025
- Organizer of various CCRG Friday lunch seminars
- Team leader of the *AsterX/CarpetX* group for the Frontier Hackathon (Virtual, June 2023)
- Referee/Reviewer for National Science Foundation, Physical Review Journals, Classical & Quantum Gravity, A&A
- Member of the American Physical Society (DGRAV, DCOMP), LISA consortium (community member), Theory & Computational Astrophysics Network, Indian Association for General Relativity & Gravitation

TEACHING & MENTORING

- Guest lecture on *Drag Forces* for undergraduate course *Classical Mechanics* (PI Prof. Hector O. Silva, UIUC)
- Guest lecture on *Harmonic Oscillators* for undergraduate course *Classical Mechanics* (PI Prof. Helvi Witek, UIUC)
- Guest lectures on *General Relativistic Magnetohydrodynamics* and *Numerical Methods for Conservation Laws* for graduate courses *Astrophysical Flows* and *Fundamentals of Theoretical Astrophysics I* (PI Prof. Manuela Campanelli, RIT)
- Designed and presented tutorials on *CarpetX* and *AsterX* during the Einstein Toolkit Workshops
- Co-mentor of five graduate students: Allen Wen (RIT, 12/23-present), Amara Nusbickel (RIT, 03/25-03/26), Maxwell Rizzo (UIUC, 08/25-present), Noora Ghadiri (UIUC, 03/26-present), Freddy Pardoe (UIUC, 03/26-present)
- Mentor of two graduate students during Einstein Toolkit hackathons: Bing-Jyun Tsao (2023), Maxwell Rizzo (2025)

TALKS

23 invited talks, seminars and tutorials, 22 contributed talks and posters.

INVITED

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| JUL 2026 | New Frontiers in Strong Gravity, Benasque, Spain
Title: <i>Relativistic outflows from compact-object binaries</i> |
| MAR 2026 | APS Global Physics Summit, Denver, USA
Title: <i>General relativistic magnetohydrodynamics on GPUs</i> |
| FEB 2026 | Nuclear Physics from Multi-Messenger Mergers Seminars, online
Title: <i>Towards end-to-end modelling of jets from BNS mergers (Seminar)</i> |
| DEC 2025 | European Einstein Toolkit Meeting, Online
Title: <i>Tutorial on the AsterX code</i> |
| OCT 2025 | Astrophysics, Relativity, and Cosmology Seminars, UIUC, USA
Title: <i>End-to-end modelling of jets from BNS mergers (Seminar)</i> |
| AUG 2025 | Einstein Toolkit CSSI workshop, UIUC, USA
Title: <i>nuX-M1: a new GPU-accelerated moment-based neutrino radiation transport code</i> |
| JUNE 2025 | North American Einstein Toolkit Workshop 2025, University of Texas, Austin, USA
Title: <i>The AsterX and nuX code</i> |
| APR 2025 | The 7th Neighbourhood workshop, Penn State University, USA
Title: <i>End-to-end modelling of jets from BNS mergers (Plenary)</i> |
| OCT 2024 | 130th Topical Symposium of the New York State Section of the APS, SUNY Polytechnic Institute, USA
Title: <i>End-to-end modelling of jets from BNS mergers</i> |
| OCT 2024 | NEOSGrav Conference, Goa, India
Title: <i>End-to-end modelling of jets from BNS mergers</i> |
| JULY 2024 | Solving the Boltzmann Equation for Neutrino Transport, ICERM, USA
Title: <i>AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes</i> |
| JULY 2024 | North American Einstein Toolkit Workshop 2024, Louisiana State University, USA
Title: <i>AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes</i> |
| AUG 2023 | INT 23-2 Workshop, University of Washington, Seattle, USA
Title: <i>Binary Neutron Star Merger Simulations: Long-Lived Remnants, Short Gamma-Ray Bursts & Kilonovae</i> |
| JULY 2023 | North American Einstein Toolkit School 2023, Rochester Institute of Technology, USA
Title: <i>Introduction to GRMHD & the AsterX code</i> |
| JULY 2023 | North American Einstein Toolkit School 2023, Rochester Institute of Technology, USA
Title: <i>The CarpetX/AsterX code (Tutorial)</i> |
| MAY 2023 | Numerical Relativity Seminars, Online
Title: <i>Binary Neutron Star Merger Simulations: Long-Lived Remnants, Short Gamma-Ray Bursts & Kilonovae</i> |

OCT 2022	EMMI-IReNA GSI Workshop, Online Title: <i>Magnetized BNS merger simulations & prospects of launching SGRB jets</i>
JUNE 2022	TCAN Meeting 2022, Online Title: <i>Binary Neutron Star Merger Simulations with WhiskyMHD and Spritz</i>
JUNE 2022	North American Einstein Toolkit Workshop, Online Title: <i>The CarpetX Framework (Tutorial)</i>
JUNE 2022	North American Einstein Toolkit Workshop, Online Title: <i>The Spritz Code: GRMHD with Neutrinos</i>
JULY 2021	TCAN Meeting 2021: BNS/BH-NS Merger Workshop Title: <i>Magnetized BNS Merger Simulations with Long Post-Merger Evolution</i>
JULY 2021	16th Marcel Grossmann Virtual Meeting Title: <i>Binary Neutron Star Merger Simulations: Long-Lived Remnant, Short Gamma-Ray Bursts, & Kilonovae</i>
JUNE 2021	ECT* Workshop: Neutron Stars as Multi-Messenger Laboratories for Dense Matter Title: <i>GRMHD Simulations of Binary Neutron Star Mergers</i>
CONTRIBUTED	
JUL 2026	UK & Ireland Discs Meeting 2026, Online Title: <i>Jet variability in small mass ratio supermassive black-hole binaries</i> (Poster)
JUL 2026	16th International LISA Symposium, College Park, USA Title: <i>Jet variability in small mass ratio supermassive black-hole binaries</i>
MAR 2026	The 3rd International Online Conference on Universe, Online Title: <i>Jet-environment interaction after delayed collapse in BNS mergers</i>
DEC 2024	Numerical Relativity Community, Online Title: <i>AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes</i>
APR 2024	APS April Meeting 2024, Sacramento, California, USA Title: <i>End-to-end modelling of jets from binary neutron star mergers</i>
APR 2023	APS April Meeting 2023, Minneapolis, Minnesota, USA Title: <i>AsterX: a new open-source GPU-accelerated GRMHD code for dynamical spacetimes</i>
JULY 2022	COSPAR – 44 th Scientific Assembly, Athens, Greece Title: <i>Magnetized BNS merger simulations & prospects of launching SGRB jets</i>
MAY 2022	PHAROS Conference 2022, Rome, Italy Title: <i>Binary Neutron Star Merger Simulations with GRMHD code Spritz</i>
APR 2022	APS April Meeting, Online Title: <i>Binary Neutron Star Merger Simulations with GRMHD code Spritz</i>
Nov 2021	IAU Online Symposium 363 – Neutron Star Astrophysics at the Crossroads Title: <i>Magnetars born in BNS mergers, Short GRBs & Kilonovae</i>
JULY 2021	North American Einstein Toolkit Summer School Title: <i>A new primitive variable recovery scheme in GRMHD code Spritz</i>
APR 2021	APS April Online Meeting Title: <i>The Spritz Code: Improving the Robustness of GRMHD</i>
FEB 2021	PHAROS-CompOSE Conference, Online Title: <i>A new primitive variable recovery scheme in GRMHD code Spritz</i>
FEB 2021	TEONGRAV Workshop, Online Title: <i>Binary Neutron Star Merger Simulations: Long-Lived Remnant, Short Gamma-Ray Bursts & Kilonovae</i>
DEC 2020	IAGRG Meeting 2020, Online Title: <i>Role of Magnetic Fields in Binary Neutron Star Merger Simulations</i>
JULY 2020	TCAN on Binary Neutron Stars Workshop 2020 Title: <i>Role of Magnetic Fields in Binary Neutron Star Merger Simulations</i>
APR 2020	APS Virtual April Meeting Title: <i>GRMHD Simulations of BNS Mergers: 100 ms of Long-Lived Remnant Evolution</i>
SEP 2019	European Einstein Toolkit Meeting, London, UK Title: <i>GRMHD Simulations of BNS Mergers: 100 ms of Long-Lived Remnant Evolution</i>
APR 2019	PHAROS Conference, Playa de Aro, Spain Title: <i>GRMHD Simulations of BNS Mergers: 100 ms of Long-Lived Remnant Evolution</i>
MAR 2019	PHAROS PhD School, University of Jena, Germany

	Title: <i>GRMHD Simulations of BNS Mergers (Poster)</i>
JAN 2019	IAGRG Meeting 2019, BITS Pilani, Hyderabad, India
	Title: <i>GRMHD Simulations of BNS Mergers (Poster)</i>
JUN 2018	AstroMundus Retreat 2018, Universitätszentrum Obergurgl, Austria
	Title: <i>GRMHD Simulations of BNS Mergers (Poster)</i>
AUG 2015	Center of Axion and Precision Physics, KAIST, South Korea
	Title: <i>Precision Particle Tracking (Poster)</i>

PUBLIC OUTREACH

MAY 2020 - PRESENT	Volunteer - Skype A Scientist Held several sessions on astronomy with classrooms/families via Skype/Zoom
APR 2023	Volunteer - <i>Imagine RIT: Creativity and Innovation Festival</i> Exhibitor: Astrophysics department
MAY 2020	Organizer - Online sessions on Let's Talk Astronomy Talk on Galaxies, Stars and Planets
APR 2020	Article contribution for Astrobites <i>Short Gamma Ray Burst Central Engines: A Curious Case of Time-Reversal</i>

COMMUNITY WORK

OCT 2023 – JUL 2025	Dog-care Volunteer (2–3 hours per week) Rochester Animal Services, Rochester, NY, USA
APR 2020 – JUN 2025	Active Listener 7Cups
JUL 2014 – SEP 2016	Ed-Support Volunteer Make A Difference (MAD), Bangalore, India
MAY 2012 – JUN 2014	Head of Operations & Co-Director Project Book-Wings, The Green Nest, Chennai, India
FEB 2012 – NOV 2013	Active Volunteer Child Rights and You (CRY), Bangalore/Chennai, India

CONTACT INFORMATION FOR ACADEMIC REFEREES

- **Prof. Helvi Witek**
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Email: hwitek@illinois.edu
Relation: current postdoctoral advisor
- **Prof. Manuela Campanelli**
Center for Computational Relativity and Gravitation & School of Mathematical Sciences,
Rochester Institute of Technology, 85 Lomb Memorial Drive, Rochester, NY 14623, USA
Email: mxcsma@rit.edu
Relation: former postdoctoral advisor
- **Dr. Riccardo Ciolfi**
INAF–Osservatorio Astronomico di Padova, Vicolo dell'Osservatorio 5, I-35122 Padova, Italy
INFN, Sezione di Padova, Via Francesco Marzolo 8, I-35131 Padova, Italy
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Relation: PhD advisor